

Ratio & Proportion (Advanced)

COMPLETE LEARNING GUIDE

What You'll Learn

Beyond the basic 'divide a total in a given ratio' questions covered in the Percentages & Ratios sub-module, placement tests probe a deeper toolkit: distinguishing direct from inverse proportion, continued proportion and mean/third proportionals, the componendo-dividendo algebraic shortcut, joint variation combining direct and inverse relationships, and ratio problems involving ages across time. The single most useful technique across all of these is the 'k-method' — representing an unknown ratio $a:b:c$ as $2k:3k:5k$ and solving for k algebraically — which converts almost any ratio word problem into a one-variable linear equation. This sub-module works through each advanced ratio pattern with a complete worked example, building the algebraic fluency needed to recognize which technique a disguised ratio question actually requires.

1. Direct and Inverse Proportion

In direct proportion, two quantities increase or decrease together, keeping their ratio constant ($x/y = k$). In inverse proportion, one quantity increases as the other decreases, keeping their PRODUCT constant ($xy = k$). Misidentifying which relationship applies is the most common error — 'more workers, fewer days' is inverse; 'more items, proportionally more cost' is direct.

- Direct: 5 kg of apples cost ₹250. Cost per kg = 50, so 8 kg costs $8 \times 50 = ₹400$.
- Inverse: 5 workers complete a task in 12 days. How many days would 15 workers take (same total work)? Since days $\propto 1/\text{workers}$, $5 \times 12 = 15 \times d \Rightarrow d = 4$ days.

2. Continued Proportion, Mean Proportional, and Third Proportional

Three quantities a, b, c are in continued proportion if $a:b = b:c$, which means $b^2 = ac$ — b is called the mean proportional between a and c . The third proportional to a and b is the value x such that $a:b = b:x$, giving $x = b^2/a$.

- Mean proportional between 4 and 9 = $\sqrt{4 \times 9} = \sqrt{36} = 6$.
- Third proportional to 4 and 8: $4:8 :: 8:x \Rightarrow x = 8^2/4 = 64/4 = 16$.

3. Componendo and Dividendo

If $a/b = c/d$, then componendo-dividendo states $(a+b)/(a-b) = (c+d)/(c-d)$. This rule is used to simplify equations involving sums and differences of square roots or fractions that would otherwise require messy algebra to isolate a variable.

- Verify with $a=8, b=2, c=12, d=3$ (both ratios equal 4): $(a+b)/(a-b) = 10/6 = 5/3$, and $(c+d)/(c-d) = 15/9 = 5/3$ — confirming the rule holds, which is how it's used to convert a hard-to-isolate equation into a simpler one.

4. The k-Method for Solving Ratio Equations

Whenever a ratio $a:b:c$ is given alongside an additional condition (a sum, a difference, or a product), represent the terms as $2k, 3k, 5k$ (matching the given ratio) and substitute into the condition to solve for k first — then multiply back to get each actual value. This turns almost any ratio word problem into simple linear algebra.

- $a:b:c = 2:3:5$ and $a+b+c = 40$. Let $a=2k, b=3k, c=5k$. Then $10k = 40 \Rightarrow k = 4$. So $a=8, b=12, c=20$.

5. Joint Variation: Combining Direct and Inverse Relationships

When a quantity y varies directly with one variable and inversely with another simultaneously, write $y = k \cdot x/z$, find the constant k from one known set of values, then use it to solve for an unknown in a different scenario.

- y varies directly with x and inversely with z . When $x=5, z=2, y=10$. Find k : $10 = k \cdot 5/2 \Rightarrow k=4$. Find y when $x=8, z=4$: $y = 4 \cdot 8/4 = 8$.

6. Ratios Involving Ages Across Time

Age-ratio problems require adding or subtracting the SAME number of years to BOTH terms of the ratio when moving to a future or past time — the ratio itself changes because the years added represent a smaller relative change to the older person's age.

- The present ages of A and B are in the ratio 3:5. After 6 years, the ratio becomes 2:3. Let $A=3x, B=5x$. Then $(3x+6)/(5x+6) = 2/3 \Rightarrow 3(3x+6) = 2(5x+6) \Rightarrow 9x+18 = 10x+12 \Rightarrow x=6$. Present ages: $A=18, B=30$.

7. Duplicate, Sub-Duplicate, and Triplicate Ratios

The duplicate ratio of $a:b$ is $a^2:b^2$; the sub-duplicate ratio is " a :" b ; the triplicate ratio is $a^3:b^3$. These are distinct from simply doubling or tripling a ratio's terms ($2a:2b$), which is not a different ratio at all — a frequent point of confusion.

- Duplicate ratio of $3:4 = 3^2:4^2 = 9:16$.
- Sub-duplicate ratio of $16:25 = "$ 16:" 25 = 4:5.

Common Pitfalls

- Setting up a direct-proportion equation for a scenario that is actually inversely proportional (or vice versa), especially in 'more workers/less time' style problems.
- Believing continued proportion means $b = ac$ instead of the correct $b^2 = ac$.
- Applying componendo-dividendo with the wrong sign, e.g. using $(a'' b)/(a+b)$ instead of $(a+b)/(a'' b)$.
- Forgetting to solve for k first in the k -method — plugging ratio terms like 2 and 3 directly into a condition instead of $2k$ and $3k$.
- Confusing the duplicate ratio ($a^2:b^2$) with simply doubling the ratio's terms ($2a:2b$) — these are entirely different operations.
- In age-ratio problems, applying the given ratio to only one term's future/past value while leaving the other term unchanged.
- Mixing up mean proportional ($'' (ac)$, the middle term of a 3-term continued proportion) with a simple average of a and c .

Key Takeaways

- Direct proportion: $x/y = \text{constant}$; inverse proportion: $xy = \text{constant}$ — identify which relationship a scenario describes before setting up the equation.
- Mean proportional between a and c = $'' (ac)$; third proportional to a , $b = b^2/a$.
- Componendo-dividendo: if $a/b=c/d$, then $(a+b)/(a'' b) = (c+d)/(c'' d)$.
- Use the k -method — write ratio terms as $2k$, $3k$, $5k$ — to convert any ratio word problem into solvable linear algebra.
- Joint variation: $y = k \cdot x/z$; find k from one known scenario, then apply it to solve for unknowns in another.
- Duplicate ratio = $a^2:b^2$, sub-duplicate = $'' a:'' b$, triplicate = $a^3:b^3$ — none of these equal simply scaling the ratio's terms.
- In age-ratio problems, add/subtract the same number of years to BOTH terms before re-forming the ratio for a different time.

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<https://www.geeksforgeeks.org/maths/ratio-and-proportion/>

IndiaBix — Ratio and Proportion Aptitude Questions

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